MACHINE LEARNING FROM A STATISTICAL PERSPECTIVE

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Can you identify the missing number?

4	6	5	□

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5	7	3	16

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3	5	4	17
5	7	3	16
4	6	6	32
7	9	5	38
1	4	7	27

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We learn the model (relationship) $Y = X_2X_3 - X_1$ from the training data (black part) to predict the output Y (here the predicted value is 26) for the future input (X_1, X_2, X_3).

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Y: Response variable, Output variable, Dependent variable.

X_1, X_2, X_3: Predictor variables, Input variables, ~~Independent variables~~ Covariates, Features, Measurement variables.

Can you identify the missing label (colour)?

X_1	X_2	Y
0.28	-0.05	C
-0.45	-0.55	A
-0.47	0.35	B
0.78	-0.83	C
-0.16	-0.94	A
-0.56	0.01	B
-0.95	-0.60	A
0.30	-0.09	C
-0.56	0.18	B
0.62	-0.99	C
0.61	0.40	A
-0.32	0.69	B
0.91	-0.33	C
-0.81	-0.81	A
0.69	0.21	A
-0.61	0.46	B
0.07	0.95	A
-0.24	0.10	B
0.66	0.24	A
0.72	-0.75	C
0.07	-0.13	□

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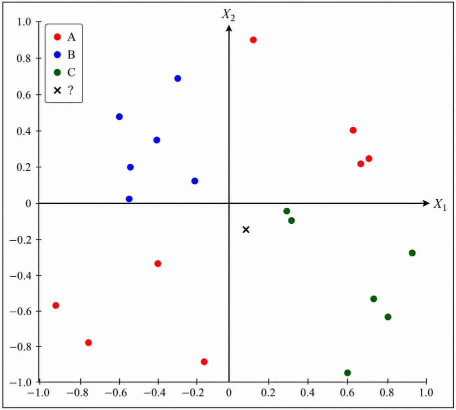
0.07	-0.13	□



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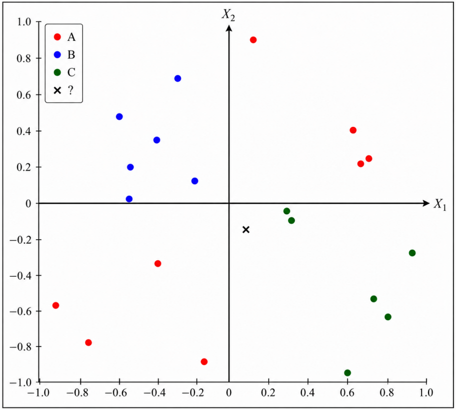


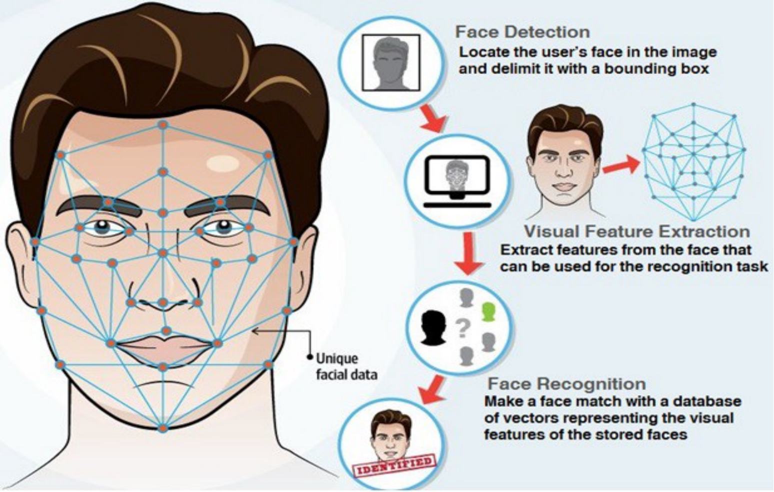
Decision rule:

$X_1 X_2 > 0$	**A**
$X_1 < 0,\ X_2 > 0$	**B**
$X_1 > 0,\ X_2 < 0$	**C**

Can you identify the person??

Can you identify the person??

Face recognition: How does the machine work?



Some examples of human learning

- A newborn learns the voices and faces of the parents.
- A toddler learns shapes, colours, digits and the alphabet.
- We decide whether to cross the road.
- We decide when to start from home to catch the train.
- Medical practitioners analyse previous data and read medical journals to learn more about a new disease.

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Learning: Study of ‘input-output’ relationship

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Learning: Study of 'input-output' relationship

Machine Learning: Development of algorithms so that machines can learn.

Statistical Machine Learning: Development of the algorithms based on statistical theories.

Different types of learning

- Supervised (Training data contains information about both input and output (target) variables)
  - Regression (Response is numerical)
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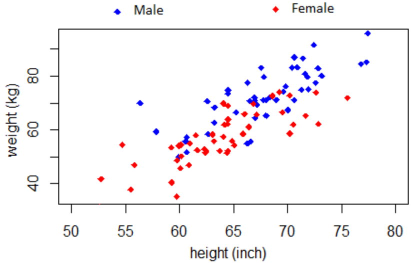
- Semi-supervised learning
- Reinforcement learning
- Transfer learning

Supervised Learning: Regression and Classification

Machine learning aims at developing algorithms for more accurate predictions based on the data.



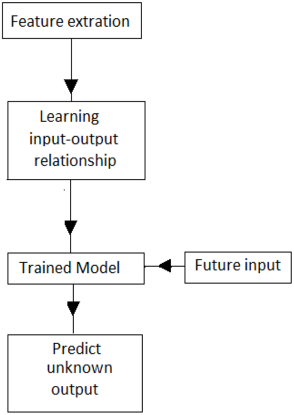
Gender	Height	Weight
Male	69	
Female	61	
	63	60
	70	74

Regression

Respose is numerical

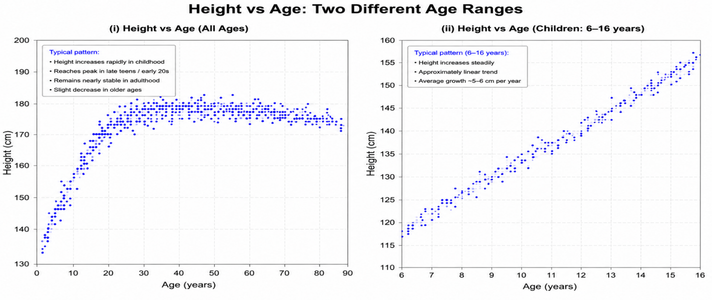
Classification

Response is cateorical



Regression

- Y: Response variable, $X_1, X_2, \ldots, X_d$: Covariates



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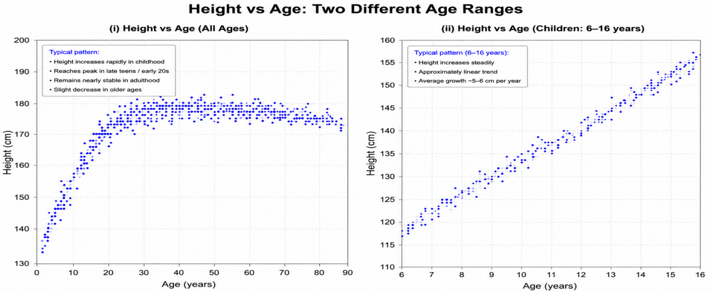

- We assume the model

$$Y = \underbrace{f(X_1, X_2, \ldots, X_d)}_{\text{signal}} + \underbrace{\epsilon}_{\text{noise}}$$

and look for estimating the function $f : \mathbb{R}^d \to \mathbb{R}$ from the data.

- Both parametric and nonparametric methods can be used for the estimation of the regression function/surface f.

Estimation of f: minimisation of expected loss

- We can define a suitable loss function $L(a, b)$ and look for minimising the expected loss $E[L(Y, \hat{f}(X_1, X_2, \ldots, X_d))]$.

Estimation of f: minimisation of expected loss

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- Squared error loss: $L(a, b) = (a - b)^2$
  - Risk$(\hat{f}) = E[(Y - \hat{f}(X_1, X_2, \ldots, X_d))^2]$ is minimised when $\hat{f}(x_1, x_2, \ldots, x_d) = E(Y \mid X_1 = x_1, X_2 = x_2, \ldots, X_d = x_d)$
  - Risk$(\hat{f}) = E[(Y - \hat{E}[\hat{f}(X_1, \ldots, X_d))^2] + Var[\hat{f}(X_1, \ldots, X_d))]$ $=$ Squared bias + Variance.

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    $=$ Squared bias + Variance.
- Absolute error loss: $L(a, b) = |a - b|$
  - $\text{Risk}(\hat{f}) = E|(Y - \hat{f}(X_1, X_2, \ldots, X_d))|$ is minimised when $\hat{f}(x_1, x_2, \ldots, x_d) = \text{Med}(Y \mid X_1 = x_1, \ldots, X_d = x_d)$
  - $\text{Risk}(\hat{f}) \leq E|Y - \text{Med}[\hat{f}(X_1, \ldots, X_d)| + MD[\hat{f}(X_1, \ldots, X_d))]$
    $=$ Absolute bias + Mean Deviation.

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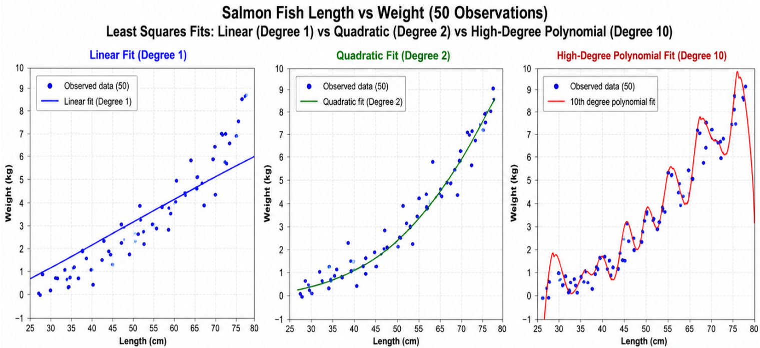
  - $\text{Risk}(\hat{f}) \leq E|Y - Med[\hat{f}(X_1, \ldots, X_d)| + MD[\hat{f}(X_1, \ldots, X_d))]$ $=$ Absolute bias + Mean Deviation.
- In practice, we minimise the emprical version of expected loss $\frac{1}{n}\sum_{\ell=1}^{n} L(y_\ell, \hat{f}(X_{1\ell}, x_{2\ell}, \ldots, x_{d\ell}))$ or $\sum_{\ell=1}^{n} L(y_\ell, \hat{f}(x_{1\ell}, x_{2\ell}, \ldots, x_{d\ell}))$

MACHINE LEARNING

Anil K. Ghosh

Bias-variance tradeoff: Underfit and overfit



Underfitting (High Bias)
The linear model cannot capture the curvature in the data.
Systematic pattern remains in residuals.

Good Fit (Balanced Bias–Variance)
Captures the nonlinear trend well without chasing the random noise.
Best generalization.

Overfitting (High Variance)
Follows the noise and random fluctuations (especially at the ends).
Poor generalization to new data.

Training Performance (Least Squares) — Linear (Degree 1)

SSE (Sum of Squared Errors)	=	69.72
R^2 (Coefficient of Determination)	=	0.646
RMSE (Root MSE)	=	1.180

Training Performance (Least Squares) — Quadratic (Degree 2)

SSE (Sum of Squared Errors)	=	24.18
R^2 (Coefficient of Determination)	=	0.880
RMSE (Root MSE)	=	0.694

Training Performance (Least Squares) — Degree 10

SSE (Sum of Squared Errors)	=	3.89
R^2 (Coefficient of Determination)	=	0.980
RMSE (Root MSE)	=	0.278

Bias–Variance Trade-off
As model complexity increases: Bias decreases but Variance increases.
The quadratic model provides the best trade-off on this dataset.

Linear regression

We assume the model $f(x_1, \ldots, x_d) = \beta_0 + \beta_1 x_1 + \cdots + \beta_d x_d$.

Data matrix:

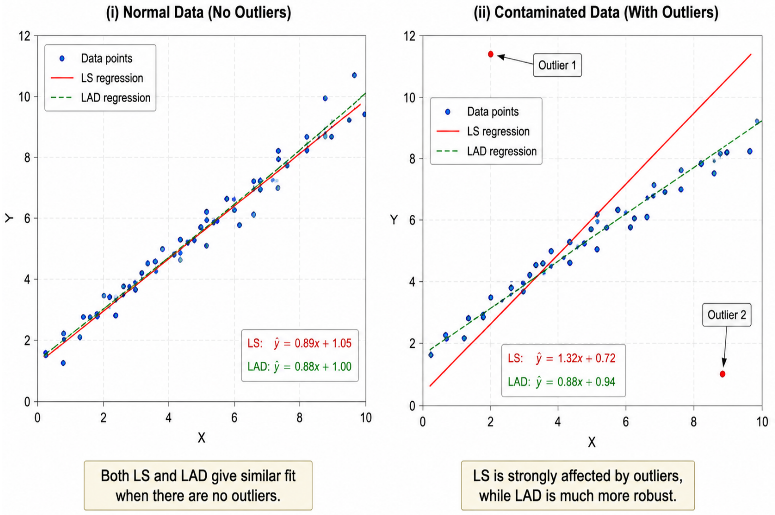
	X_1	X_2	$\cdots$	X_d	Y
1	x_{11}	x_{21}	$\cdots$	x_{d1}	y_1
2	x_{12}	x_{22}	$\cdots$	x_{d2}	y_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
ℓ	$x_{1\ell}$	$x_{2\ell}$	$\cdots$	$x_{d\ell}$	y_ℓ
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
n	x_{1n}	x_{2n}	$\cdots$	x_{dn}	y_n

We minimize

$$\sum_{\ell=1}^{n} (y_\ell - \beta_0 - \beta_1 x_{1\ell} - \cdots - \beta_d x_{d\ell})^2$$: least square (LS) regression

$$\sum_{\ell=1}^{n} |y_\ell - \beta_0 - \beta_1 x_{1\ell} - \cdots - \beta_d x_{d\ell}|$$: least absolute dev. (LAD) regression

LS and LAD regression



LAD regression: Estimation of parameters

Using the following facts, one can develop an iterative algorithm to minimise $\sum_{\ell=1}^{n} |y_\ell - \beta_0 - \beta_1 x_{1\ell} - \cdots - \beta_d x_{d\ell}|$

- Fact 1: $\sum_{\ell=1}^{n} |z_\ell - \alpha_0|$ is minimised when α_0 is the median of $\{z_1, z_2, \ldots, z_n\}$.
- Fact 2: $\sum_{\ell=1}^{n} |z_\ell - \alpha w_\ell| = \sum_{\ell=1}^{n} |w_\ell| \big|(z_\ell / w_\ell) - \alpha\big|$ is minimized when α is the median of the following frequency distribution

Value	z_1/w_1	z_2/w_2	$\cdots$	z_ℓ/w_ℓ	$\cdots$	z_n/w_n
Frequency	$\|w_1\|$	$\|w_2\|$	$\cdots$	$\|w_\ell\|$	$\cdots$	$\|w_n\|$

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Another option: **iteratively re-weighted least squares** (IRLS) algorithm, where one starts with $\boldsymbol{\beta}^{(1)} = (\beta_0^{(1)}, \beta_1^{(1)}, \ldots, \beta_d^{(1)})$ and at the $t+1$-th ($t \geq 1$) step, gets $\boldsymbol{\beta}^{(t+1)}$ by minimising $\psi_t(\boldsymbol{\beta}) = \sum_{\ell=1}^{n} w_\ell^{(t)} \left(y_\ell - \mathbf{x}_\ell^\top \boldsymbol{\beta}\right)^2$, where $w_\ell^{(t)} = \left|y_\ell - \mathbf{x}_\ell^\top \boldsymbol{\beta}^{(t)}\right|^{-1}$. and $\mathbf{x}_\ell = (1, x_{1\ell}, x_{1\ell}, \ldots, x_{d\ell})$.

LS Regression: Estimation of Parameters

Define

$$\mathbb{X} = \begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{d1} \\ 1 & x_{12} & x_{22} & \cdots & x_{d2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1\ell} & x_{2\ell} & \cdots & x_{d\ell} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & x_{2n} & \cdots & x_{dn} \end{bmatrix}, \quad \mathbb{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_\ell \\ \vdots \\ y_n \end{bmatrix} \text{ and } \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \vdots \\ \beta_d \end{bmatrix}.$$

Minimizing $\sum_{\ell=1}^{n}(y_\ell - \beta_0 + \beta_1 x_{1\ell} + \cdots + \beta_d x_{d\ell})^2 = \|\mathbb{Y} - \mathbb{X}\boldsymbol{\beta}\|^2$,

one gets the least square estimate $\hat{\boldsymbol{\beta}}_{LS} = (\mathbb{X}^\top\mathbb{X})^{-1}\mathbb{X}^\top\mathbb{Y}$.

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Under standard model assumptions, we have

- $Disp(\hat{\boldsymbol{\beta}}_{LS}) = \sigma^2(\mathbb{X}^\top\mathbb{X})^{-1}$.
- $E\|\hat{\boldsymbol{\beta}}_{LS} - \boldsymbol{\beta}\|^2 = \sum_{k=0}^{d} \mathrm{Var}(\hat{\beta}_k) = \sigma^2 \sum_k (1/\lambda_k)$, where the λ_k's are the eigenvalues of $\mathbb{X}^\top\mathbb{X}$

The problem of multicollinraity: An example

Consider a problem with two covariates where X_1, X_2 and Y are standardised (each of them has mean 0 and variance 1)

- No need to consider the intercept term β_0 here and one can consider the linear model $Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon$.
- Here $\mathbb{X}^\top \mathbb{X} = n \begin{bmatrix} 1 & r \\ r & 1 \end{bmatrix}$ and $\mathbb{X}^\top \mathbb{Y} = n \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$, where $r = \text{corr}(X_1, X_2)$, $r_1 = \text{corr}(Y, X_1)$ and $r_2 = \text{corr}(Y, X_2)$.
- $\hat{\boldsymbol{\beta}}_{LS} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \frac{1}{1-r^2} \begin{bmatrix} r_1 - r\, r_2 \\ r_2 - r\, r_1 \end{bmatrix}$.
- $\text{Var}(\hat{\beta}_1) = \text{Var}(\hat{\beta}_2) = \sigma^2/[n(1-r^2)]$.

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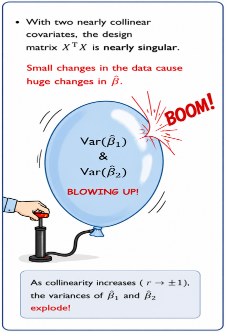
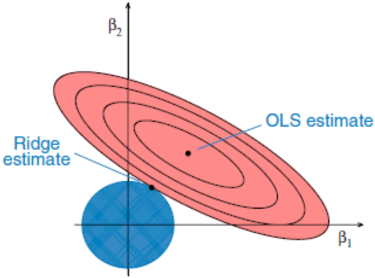
- $\text{Var}(\hat{\beta}_1) = \text{Var}(\hat{\beta}_2) = \sigma^2/[n(1-r^2)]$.
- If $r \approx \pm 1$,
    1. $|\hat{\beta}_1|$ and $|\hat{\beta}_2|$ become large.
    2. Signs of $\hat{\beta}_1$ and $\hat{\beta}_2$ can be misleading.
    3. These estimates become less reliable (high variance).



Ridge regression

We minimize $\|\mathbb{Y} - \mathbb{X}\boldsymbol{\beta}\|^2$ subject to $\|\boldsymbol{\beta}\|^2 \le C$ for some $C > 0$.

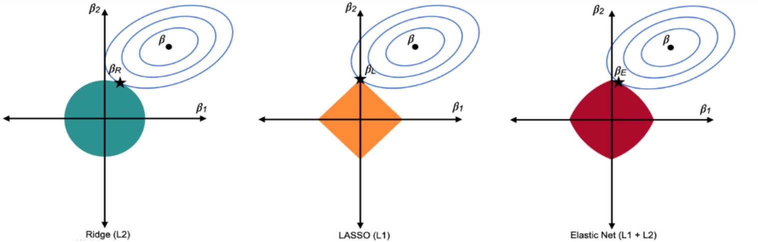


$$\|\mathbb{Y} - \mathbb{X}\boldsymbol{\beta}\|^2 = \|\mathbb{Y} - \mathbb{X}\hat{\boldsymbol{\beta}}_{LS}\|^2 + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{LS})^\top \mathbb{X}^\top \mathbb{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_{LS}).$$

- This minimisation is equivalent to the minimisation of $\|\mathbb{Y} - \mathbb{X}\boldsymbol{\beta}\|^2 + \lambda(|\boldsymbol{\beta}|^2 - C)$ for some $\lambda > 0$.
- Ridge solution: $\hat{\boldsymbol{\beta}}_R = (\mathbb{X}^\top \mathbb{X} + \lambda\,\mathbf{I})^{-1}\mathbb{X}^\top \mathbb{Y}$.

LASSO regression

We minimize $\|\mathbb{Y} - \mathbb{X}\beta\|^2$ subject to $\|\beta\|_1 \leq C$ for some $C > 0$, or equivalently, minimize $\|\mathbb{Y} - \mathbb{X}\beta\|^2 + \lambda(\|\beta\|_1 - C)$ for some $\lambda > 0$.



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Disadvantage: No closed-form expression. Usually, the coordinate-descent algorithm is used for minimisation.

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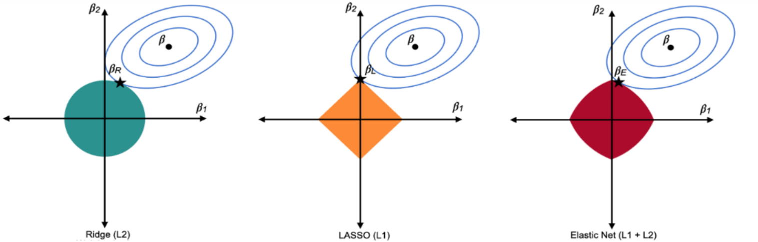

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Elastic net: considers a combination of ℓ_1 and ℓ_2 penalties. It minimizes $\|\mathbb{Y} - \mathbb{X}\boldsymbol{\beta}\|^2 + \lambda(\frac{\alpha}{2}\|\boldsymbol{\beta}\|^2 + (1-\alpha)\|\boldsymbol{\beta}\|_1),\ \lambda > 0, \alpha \in [0.1]$.

Nonparametric regression

- Piecewise constant function (naive method)
- Piecewise linear/quadratic/cubic functions (splines)
- Additive models
- Projection pursuit regression
- Artificial Neural Networks (Perceptron models)
- Generalised additive models
- Regression trees
- MARS (Multivariate Adaptive Regression Splines)
- Nearest neighbour regression
- LOESS regression
- Kernel regression
- Local linear and local polynomial regression
- Support vector regression
- Wavelets regression

Classification problem: An example

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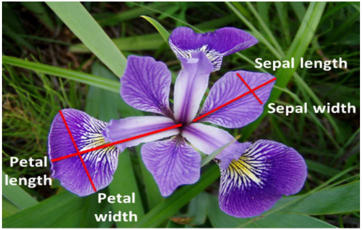
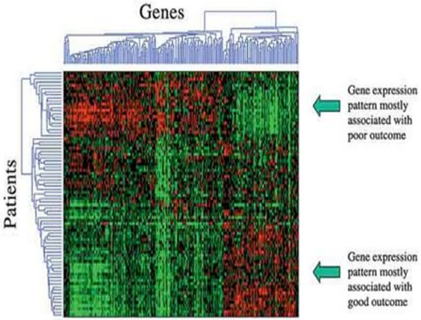
- Iris data: Four measurements are taken on each Iris flower.



X=(sepal length, sepal width, petal length, petal width). We need to identify the type of an Iris flower based on the feature vector **X**.

Classification problem: Another example

- Breast cancer data: For each patient (normal patient and cancer patients), expression levels are observed for 6137 genes



X= microarray expression levels of 6137 genes.
We need to classify a new patient based on the feature vector **X**.

Classification: mathematical framework

Data: Class-1: $\mathbf{X}_{11}, \mathbf{X}_{12}, \ldots, \mathbf{X}_{1n_1}$
Class-2: $\mathbf{X}_{21}, \mathbf{X}_{22}, \ldots, \mathbf{X}_{2n_2}$
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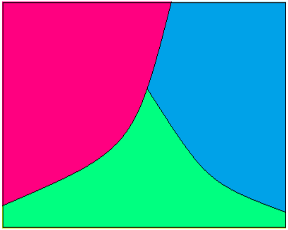
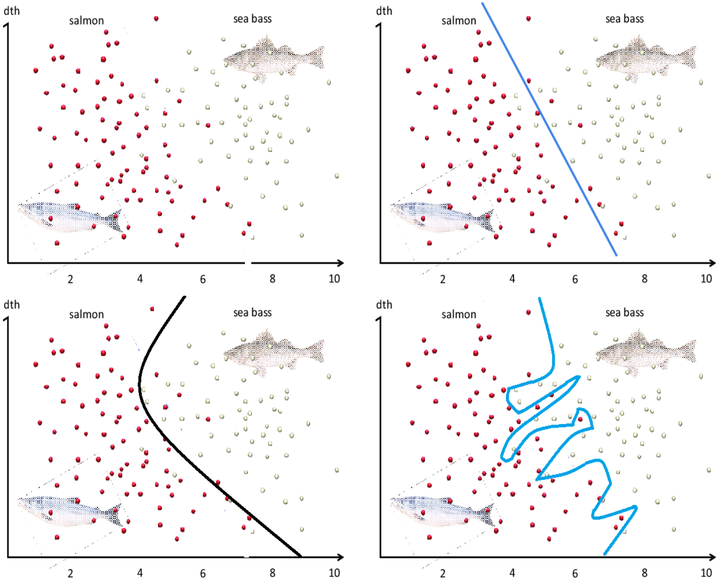


Linear and nonlinear classifiers



Loss function in classification

Consider a J-class problem,

- The response variable Y takes values in the set $\{1, 2, \ldots, J\}$.
- Decision rule: $\delta(X_1, X_2, \ldots, X_d) : \mathbb{R}^d \to \{1, 2, \ldots, J\}$.

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- Expected loss: $\text{Risk}(\delta) = E[L(\delta(X_1, X_2, \ldots, X_d), Y)]$
  $= P(\delta(X_1, X_2, \ldots, X_d) \neq Y)$
- Empirical version: $\frac{1}{n}\sum_{i=1}^{n} \mathbb{I}\{\delta(x_{1i}, x_{2i}, \ldots, x_{id}) \neq y_i\}$,
  the proportion of misclassification in the training set, is not a convex function.

How to form the partition?: An illustrative example (Example 1)

In a bolt factory, Machine-1 and Machine-2 manufacture 40% and 60% of total bolts. Of their outputs, 20% and 10%, respectively, are defective bolts. In a pack of 20 bolts manufactured by the same machine, 3 are found to be defective. Which machine is more likely to manufacture the bolts in the pack?

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  - $X \mid M_1 \sim Bin(20, 0.2)$
    $f_1(x) = \binom{20}{x}(0.2)^x(0.8)^{20-x}$ for $x = 0, 1, 2, \ldots, 20$.
  - $X \mid M_1 \sim Bin(20, 0.1)$
    $f_2(x) = \binom{20}{x}(0.1)^x(0.9)^{20-x}$ for $x = 0, 1, 2, \ldots, 20$.

Example-1 (contd.)

- Bayes theorem:-

$$p(1 \mid x) = P(M_1 \mid X = x) = \frac{P(M_1)P(X=x|M_1)}{\sum_{i=1}^{2} P(M_i)P(X=x|M_i)}$$
$$= \pi_1 f_1(x) / \sum_{i=1}^{2} \pi_i f_i(x).$$

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$$\boxed{p(1 \mid x) \geq p(2 \mid x) \Leftrightarrow \pi_1 f_1(x) \geq \pi_2 f_2(x).}$$

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- Decision: Here $x = 3$, $p(1 \mid x) = 0.419$ and $p(2 \mid x) = 0.581$. The bolts in the pack are more likely to be from Machine-2.

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x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$\delta(x)$	2	2	2	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The decision rule derived from the Bayes theorem (Bayes rule) δ forms a partition of the feature space.
$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}.$

Example-2

A bulb factory manufactures two types of electric bulbs. The lifetime of a bulb follows an exponential distribution with mean 2000 and 3000 hours for Type-1 and Type-2, respectively. Two-third of the total production are of Type-1 and the rest are of Type-2. A person bought two bulbs of the same type and their lifetimes were recorded as 2400 hours and 3600 hours, respectively. Can you guess the type of the bulbs bought by the person?

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- $\mathbf{X} = (X_1, X_2)$: Feature vector (lifetimes of the two bulbs)
- Population distributions (probability density functions):-
  - $X_1, X_2 \mid T_1 \overset{iid}{\sim} \exp(2000)$
    $f_1(\mathbf{x}) = f_1(x_1, x_2) = \left(\frac{1}{2000}\right)^2 e^{-(x_1+x_2)/2000}$ for $x_1, x_2 \geq 0$.
  - $X_1, X_2 \mid T_2 \overset{iid}{\sim} \exp(3000)$
    $f_2(\mathbf{x}) = f_2(x_1, x_2) = \left(\frac{1}{3000}\right)^2 e^{-(x_1+x_2)/3000}$ for $x_1, x_2 \geq 0$.

Example-2 (contd.)

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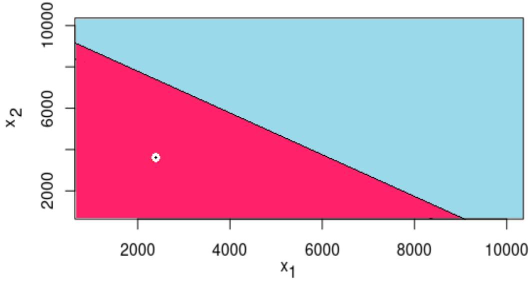
- Decision: Here $\mathbf{x} = (x_1, x_2) = (2400, 3600)$, $x_1 + x_2 = 6000$. It is more likely that the bulbs are of Type-1.



Bayes rule forms a linear partition of the feature space.

How to find the best classifier (best partition) ?

Consider a two class problem.

π_1 and π_2: prior probabilities.

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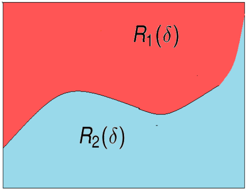
Decision (δ)	Actual: Class-1	Actual: Class-2
Class-1	OK	Error-2
Class-2	Error-1	OK



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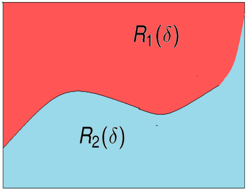

Expected loss (misclassification probability) of the decision rule (classifier) δ is given by

$$\text{Risk}(\delta) = \pi_1 \int_{R_2(\delta)} f_1(\mathbf{x})d\mathbf{x} + \pi_2 \int_{R_1(\delta)} f_2(\mathbf{x})d\mathbf{x}$$

$$= \pi_1 + \int_{R_1(\delta)} [\pi_2 f_2(\mathbf{x}) - \pi_1 f_1(\mathbf{x})]d\mathbf{x}$$

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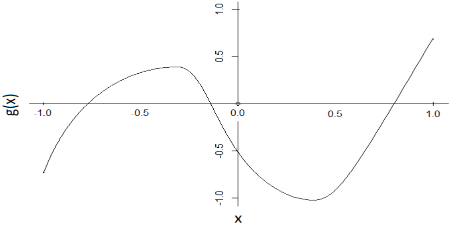

We know that $\int_A g(\mathbf{x})d\mathbf{x}$ is minimum when $A = \{\mathbf{x} : g(\mathbf{x}) < 0\}$.

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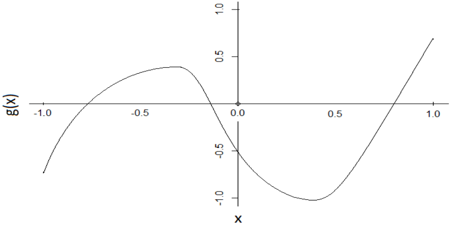

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- $\mathcal{E}(\delta) = \pi_1 + \int_{R_1(\delta)} [\pi_2 f_2(\mathbf{x}) - \pi_1 f_1(\mathbf{x})]d\mathbf{x}$ will be minimum if

  $R_1(\delta) = \{\mathbf{x} : \pi_1 f_1(\mathbf{x}) > \pi_2 f_2(\mathbf{x})\}$, i.e., $\mathbf{x}$ is classified to Class-1

  if $\pi_1 f_1(\mathbf{x}) > \pi_2 f_2(\mathbf{x})$ or $p(1 \mid \mathbf{x}) > p(2 \mid \mathbf{x})$.

- The decision rule δ thus obtained is called the Bayes rule/ Bayes classifier/ Oracle Classifier.

Bayes classifier

- Notations:-

  π_j: prior probability of the j-th class ($j = 1, 2, \ldots, J$).

  $f_j(\cdot)$: density/mass function of the j-th class.

  $p(j \mid \mathbf{x})$: posterior probability of the j-th class given an observation $\mathbf{x}$.

- Bayes (Oracle) classifier :-

$$\delta(\mathbf{x}) = j \text{ if } p(j \mid \mathbf{x}) > p(i \mid \mathbf{x}) \text{ for all } i \neq j \text{ or}$$
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- Bayes rule can't be used in practice.

- We need to estimate the posterior probabilities or the density/mass functions from the data.

Some popular classification methods

- Classification using estimates of density functions
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- Other classifiers
  - Support vector machine (SVM)

    It assumes a specific parametric form of the separating surface and then optimizes a suitable objective function to estimate the associated parameters.

Classification under regression framework

Consider a two-class classification problem. Define $Y = 1$ if the observation comes from the first class, and 0 otherwise.

Note that we can classify $\mathbf{x}$ to class-1 if $E(Y \mid \mathbf{x}) = p(1 \mid \mathbf{x}) > 0.5$. So, one can regression Y on $\mathbf{X}$ to estimate $p(1 \mid \mathbf{x})$.

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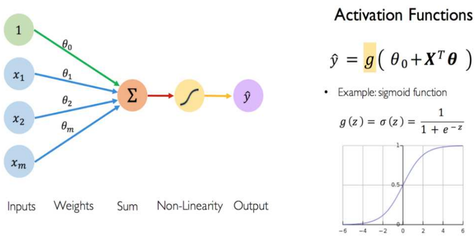
For J-class problem, define J indicators, one for each class.

ANN: Single-layer and Multi-layer perceptrons



Single-layer perceptron model leads to linear classification.

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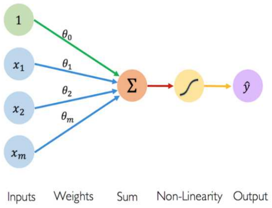
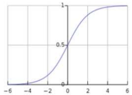


Activation Functions

$$\hat{y} = g\left(\theta_0 + X^T \boldsymbol{\theta} \right)$$

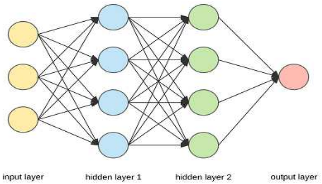
- Example: sigmoid function

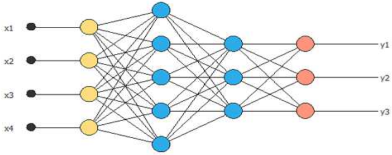
$$g(z) = \sigma(z) = \frac{1}{1 + e^{-z}}$$



Single-layer perceptron model leads to linear classification.

For estimating more complex class boundaries, we use multi-layer perceptrons with one or more hidden layers.





References

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